
Mohammed Ehab

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Projects Page: mohammedehab2002.github.io/projects

Education

Massachusetts Institute of Technology (MIT) (2021-2026)

BS/MS in Electrical Engineering and Computer Science, minor in Mathematics.

GPA: 5.0

Graduate Focus: RL fine-tuning of LLMs and diffusion models. Researching methods to shorten LLM Chains-of-Thought and to improve the designability of protein diffusion models.

Notable Courses: Deep Learning, Diffusion Models, RL, Underactuated Robotics, Autonomous Vehicles, Bio-Inspired Robotics, Distributed Algorithms, and Multi-Agent Learning.

Notable Skills:-

Languages: Python, C++, Julia, MATLAB.

Frameworks: PyTorch, Git, ROS2, GTSAM, Pandas.

Research Experience

MIT SPARK Lab | Robotics Researcher

January 2025-May 2025

Integrated recent deep-learning advances in vision (SALAD, RoMa) into the SPARK lab SLAM pipeline. Developed rigorous statistical models for loop closures, leading to a 30% decrease in pose estimation error.

Jump Trading | Quantitative Researcher

July 2024-September 2024

Developed pricing models and execution algorithms for the firm's algorithmic trading strategies.

MIT Robot Locomotion Group | Robotics Researcher

July 2023-June 2024

Led the work on an algorithm for approximating the reachability sets of underactuated robotic systems using SOS programming, with the goal of designing fast agile controllers and certifying safety.

MIT Computational Web Group | ML Researcher

December 2022-February 2023

Worked on a framework for distributed machine learning that outperforms federated and even centralized learning with a 5% increase in accuracy over the CIFAR-10 dataset. Results accepted into the ICML workshop on localized learning.

MIT Computational Connectomics Group | ML Researcher

June 2022-August 2022

Studied adversarial robustness of deep neural networks; explored GANs and diffusion models to project adversarial examples back into distribution.

Other Experience

Competitive Programming (2015-2021)

- 2× Silver, 1× Bronze medalist, International Olympiad in Informatics (IOI). Codeforces Grandmaster (top 1%).
- Created more than 70 novel problems for 20000+ participants.
- Collaborated in a team of 5 to run the Egyptian Olympiad in Informatics.